CSE 4137

Machine Learning
Module-06
Genetic Algorithm



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Week-12

- Introduction
- Implementation

Genetic Algorithm

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What is Genetic Algorithm?

A genetic algorithm is an adaptive heuristic search algorithm inspired by "Darwin's theory of evolution in Nature." It is used to solve optimization problems in machine learning. It is one of the important algorithms as it helps solve complex problems that would take a long time to solve.
Genetic Algorithms are being widely used in different real-world applications, for example, **Designing electronic circuits, code-breaking, image processing, and artificial creativity.**

Foundation of Genetic Algorithm

Genetic algorithms are based on an analogy with the genetic structure and behavior of chromosomes of the population. Following is the foundation of GAs based on this analogy –
1.Individuals in the population compete for resources and mate
2.Those individuals who are successful (fittest) then mate to create more offspring than others
3.Genes from the “fittest” parent propagate throughout the generation, that is sometimes parents create offspring which is better than either parent.
4.Thus each successive generation is more suited for their environment.

Phases of Genetic Algorithm

- Initialization
- Fitness Assignment
- Selection
- Crossover (Reproduction)
- Termination

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1. **Initialization**
- The process of a genetic algorithm starts by generating the set of individuals, which is called population (**A1-A4**).
- Here each individual (A1-A4) is the solution for the given problem.
- An individual contains or is characterized by a set of parameters called Genes.
- Genes are combined into a string and generate chromosomes, which is the solution to the problem.
- One of the most popular techniques for initialization is the use of random binary strings.

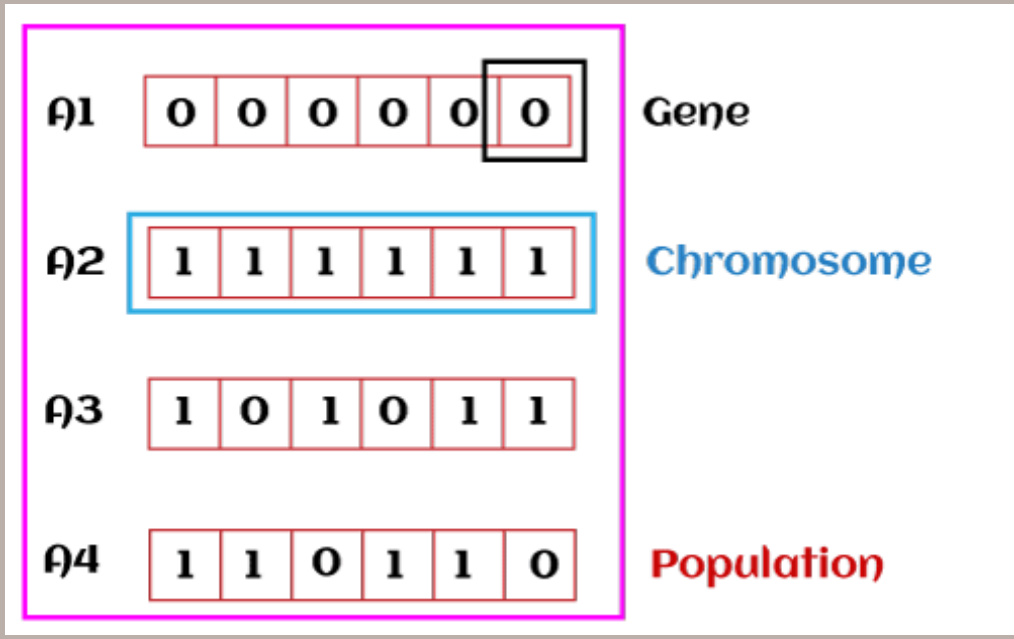
2. Fitness Score

A Fitness Score is given to each individual which shows the ability of an individual to “compete”. The individual having optimal fitness score (or near optimal) are sought.

3 Selection phase

involves the selection of individuals for the reproduction of offspring. All the selected individuals are then arranged in a pair of two to increase reproduction. Then these individuals transfer their genes to the next generation.

There are three types of Selection methods available, which are:
- Roulette wheel selection
- Tournament selection
- Rank-based selection



•**Population:** Population is the subset of all possible or probable solutions, which can solve the given problem.

•**Chromosomes:** A chromosome is one of the solutions in the population for the given problem, and the collection of gene generate a chromosome.

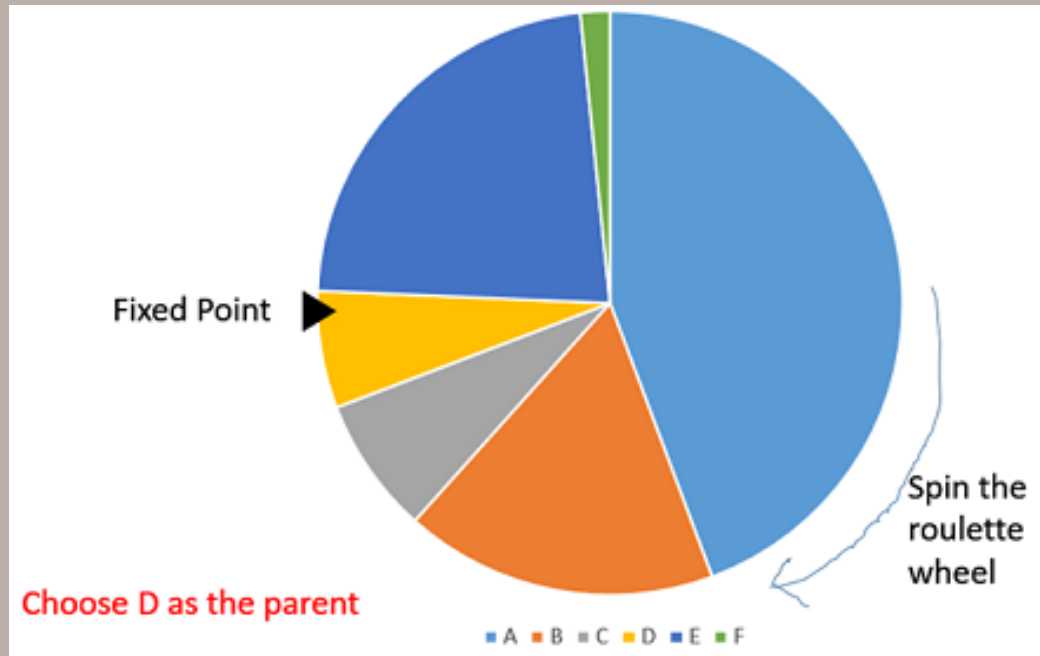
•**Gene:** A chromosome is divided into a different gene, or it is an element of the chromosome.

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3. 1 Roulette wheel selection

- Fitness Proportionate Selection is one of the most popular ways of parent selection.
- In this method, individual can become a parent with a probability which is proportional to its fitness.
- Therefore, fitter individuals have a higher chance of selection and propagating their features to the next generation.



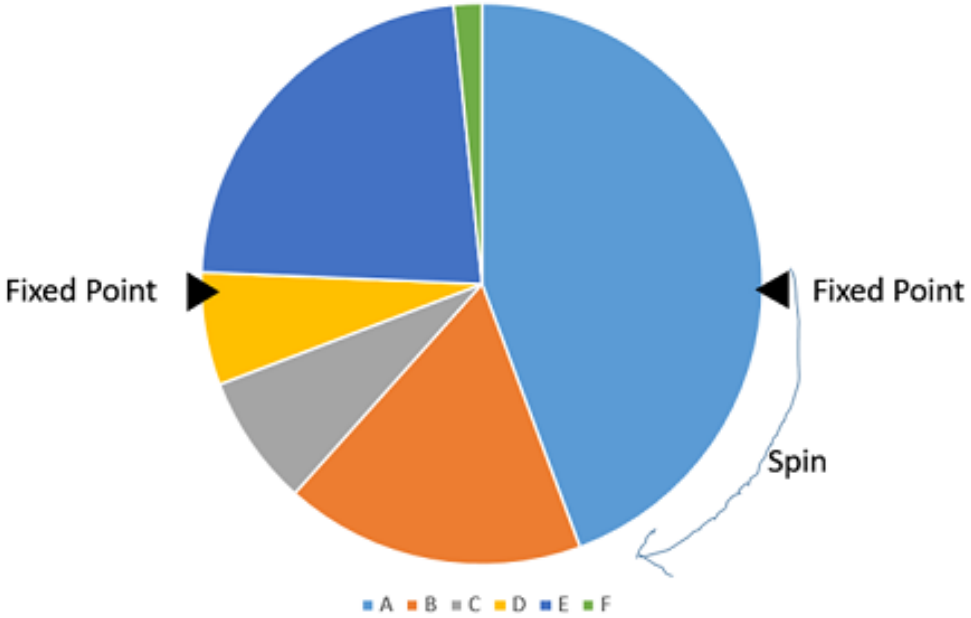
Chromosome	Fitness Value
A	8.2
B	3.2
C	1.4
D	1.2
E	4.2
F	0.3

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3. 2 Stochastic Universal Sampling (SUS)

- Stochastic Universal Sampling is quite similar to Roulette wheel selection, however instead of having just one fixed point, we have multiple fixed points as shown in the following image.
- Therefore, all the parents are chosen in just one spin of the wheel.
- Also, such a setup encourages the highly fit individuals to be chosen at least once.



Chromosome	Fitness Value
A	8.2
B	3.2
C	1.4
D	1.2
E	4.2
F	0.3

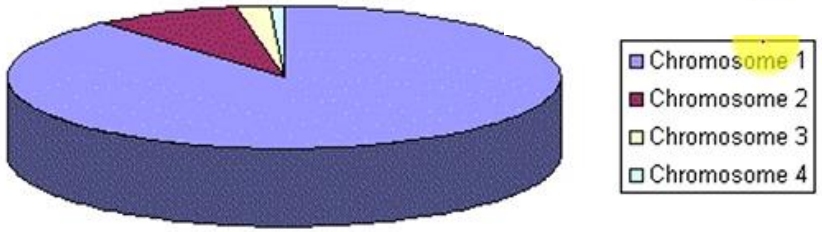
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3. 3 Ranking Selection

Problems of Roulete Selection

- The previous selection will have problems when the fitnesses differs very much.
- For example, if the best chromosome fitness is 90% of all the roulette wheel then the other chromosomes will have very few chances to be selected.



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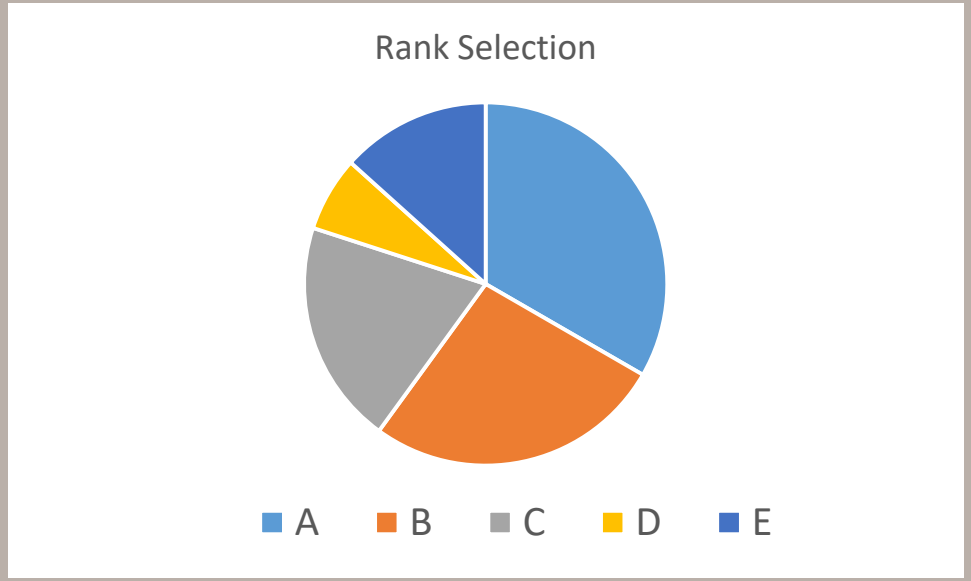
3. 3 Ranking Selection

In Roulette wheel the chromosome with high fitness value has great chance to be selected always. To overcome this selection process is not based on rank rather than fitness. Each individual is given a rank based on their fitness. The low fitness gets higher rank.

Chrom some	Fitness	Roulete	Rank	Rank Select
A	5	50%	5	33.3
B	2	20%	4	26.7
C	1.5	15%	3	20
D	0.5	5%	1	6.67
E	1	10%	2	13.3

Total rank value=15

% of rank selection=rank/(total rank)*100



Here chance of chromosome with high fitness is getting decreased and have lower fitness valued chromosome is getting increased

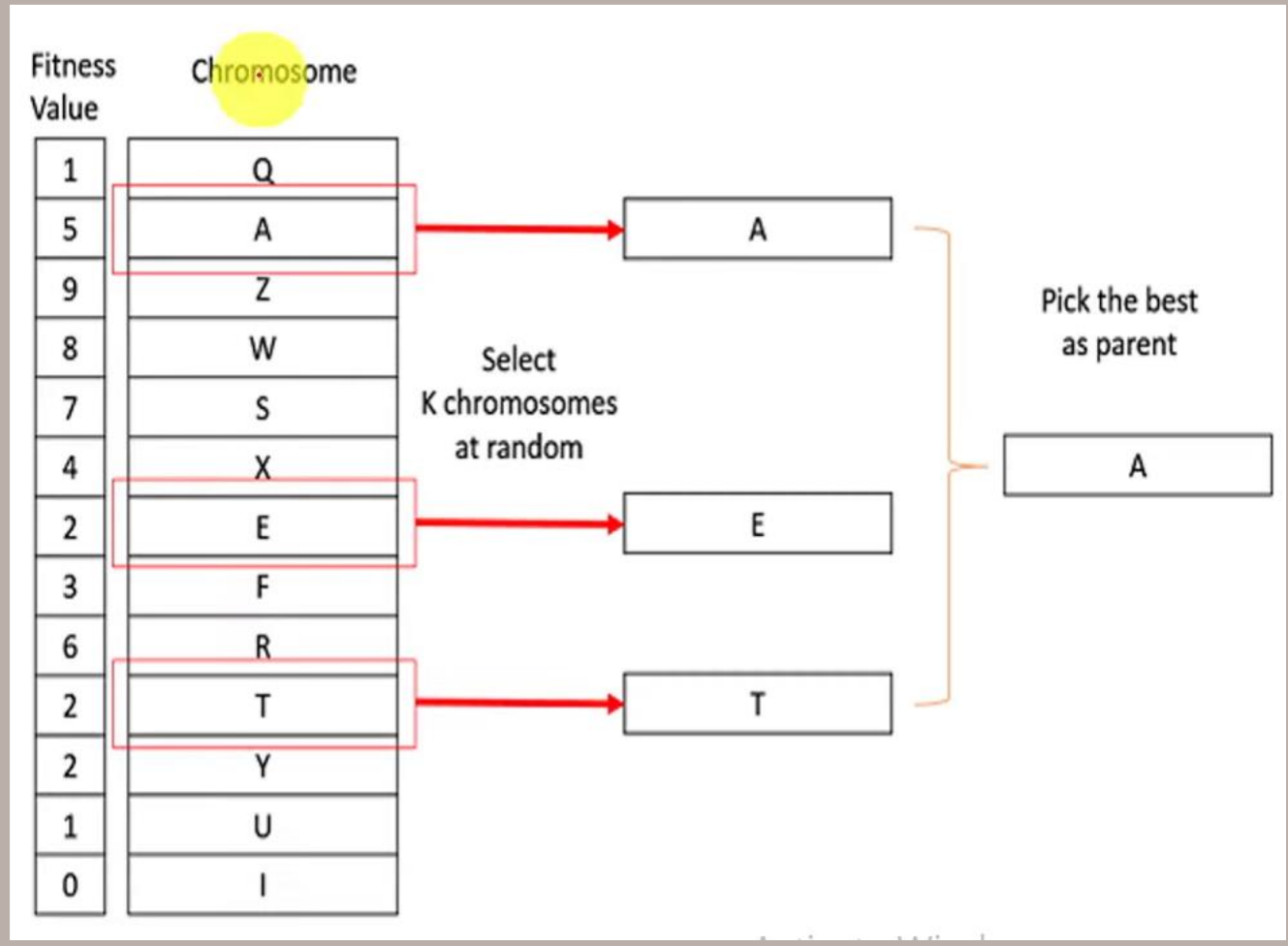
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3. 4 Tournament Selection

- In K-Way tournament selection, we select K individuals from the population at random and select the best out of these to become a parent.
- The same process is repeated for selecting the next parent.

Here K=3



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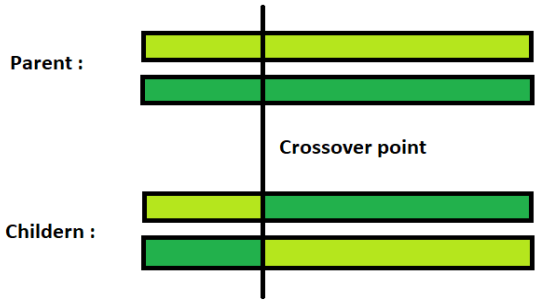
4. **Crossover**

After the selection process, the creation of a child occurs in the reproduction step. In this step, the genetic algorithm uses two variation operators that are applied to the parent population.

- **Types of Crossover**

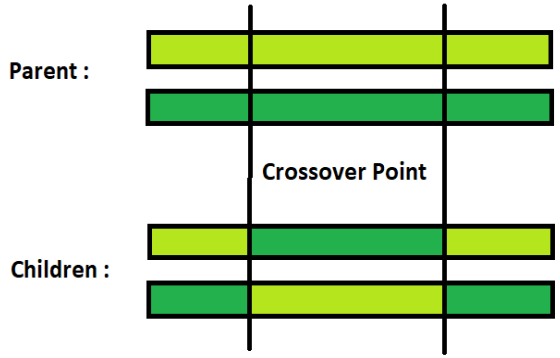
  Single Point Crossover: A crossover point on the parent organism string is selected. All data beyond that point in the organism string is swapped between the two parent organisms

  Two-Point Crossover : This is a specific case of a N-point Crossover technique. Two random points are chosen on the individual chromosomes (strings) and the genetic material is exchanged at these points.



Chromosome1	11011|00100110110
Chromosome2	11011|11000011110
Offspring1	11011|11000011110
Offspring2	11011|00100110110

Single Point Crossover



Chromosome1	11011|00100|110110
Chromosome2	10101|11000|011110
Offspring1	11011|11000|110110
Offspring2	10101|00100|011110

Two Point Crossover

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5. **Mutation**

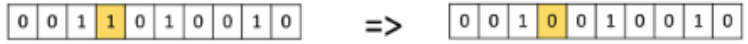
In simple terms, mutation may be defined as a small random tweak in the chromosome, to get a new. Sometimes we use mutation probability mp to do the mutation of the offspring

Offspring	1	0	0	1	1
Mutation Prob	0	1	0	0	1
Muted Offspring	1	1	0	1	0

Here flip the bits of chromosome if mutation probability is 1 others no change

Bit Flip Mutation

In this bit flip mutation, we select one or more random bits and flip them. This is used for binary encoded GAs.

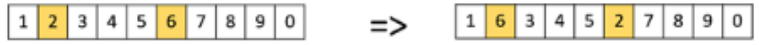


Random Resetting

Random Resetting is an extension of the bit flip for the integer representation. In this, a random value from the set of permissible values is assigned to a randomly chosen gene.

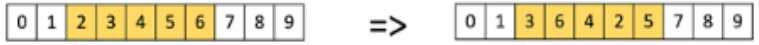
Swap Mutation

In swap mutation, we select two positions on the chromosome at random, and interchange the values. This is common in permutation based encodings.



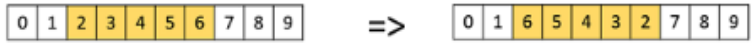
Scramble Mutation

Scramble mutation is also popular with permutation representations. In this, from the entire chromosome, a subset of genes is chosen and their values are scrambled or shuffled randomly.



Inversion Mutation

In inversion mutation, we select a subset of genes like in scramble mutation, but instead of shuffling the subset, we merely invert the entire string in the subset.



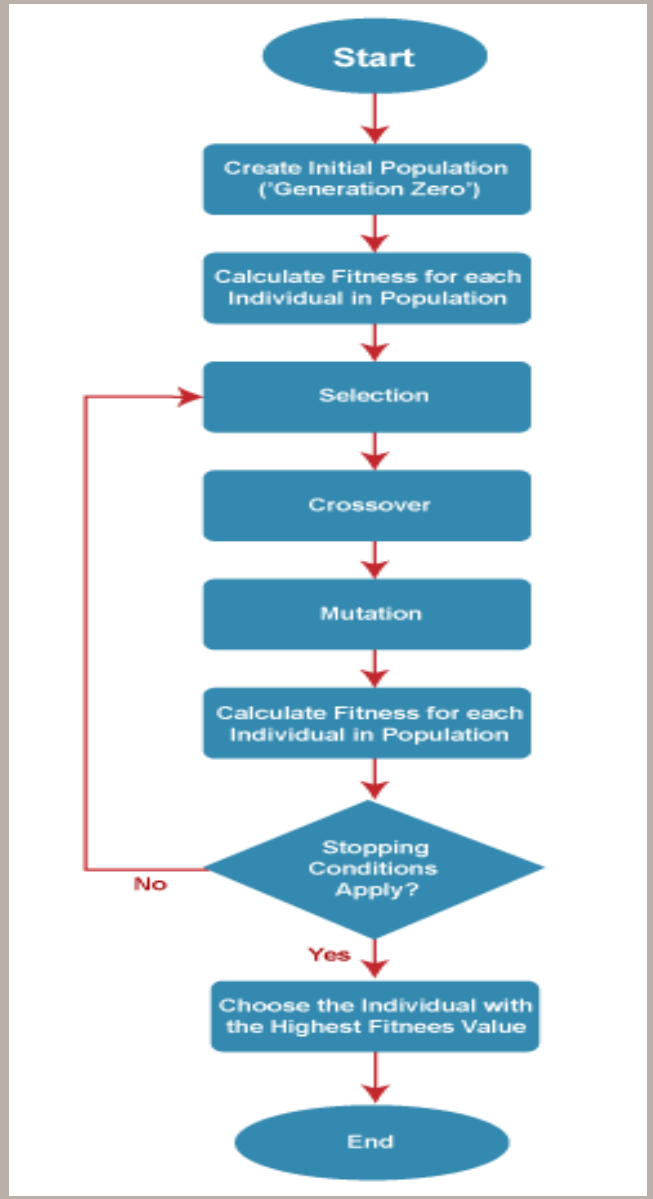
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6. Termination

After the reproduction phase, a stopping criterion is applied as a base for termination. The algorithm terminates after the threshold fitness solution is reached. It will identify the final solution as the best solution in the population.

General Workflow of a Simple Genetic Algorithm



•Solved Example

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Problem

Maximize the function $f(x)=x^2$, where x value range from 0-31

Step 1- Choose an encoding technique, a selection operator, and a crossover operator
Step 2- Choose a population size
Step 3- Randomly choose the initial population
Step 4- Select parental chromosomes
Step 5- Perform Crossover
Step 6- Evaluation of offsprings
Step 7- Repeat the process

Solution:

Step 1-
•**Encoding technique-** Binary encoding
•**Selection operator-** Roulette Wheel Selection
•**Crossover operator-** Single point crossover
Step 2-
Population size (n) = 4
Step 3-
Initial population (x value) = 12, 25, 5, 19
Step 4-

String No.	Initial Population (Randomly Selected)	X Value	Fitness $f(x) = x^2$	Prob	% Prob	Expected Count	Actual Count
1	01100	12	144	0.1247	12.47	0.4987	1
2	11001	25	625	0.5411	54.11	2.1645	2
3	00101	5	25	0.0216	2.16	0.0866	0
4	10011	19	181	0.3126	31.26	1.2502	1
Sum			1155	1.0	100	4	4
Average			288.75	0.25	25	1	1
Maximum			625	0.5411	54.11	2.1645	2

$$Prob = \frac{f(x)}{\sum f(x)}$$

$$Expected\ Count = \frac{f(x_i)}{Avg(\sum f(x))}$$

$$Actual\ Count = Round(Expected\ Count)$$

Now How to select chromosome for next steps?
Look at the actual count and exclude 5 (chromosome) as its count value=0
Based on count values take 12 and 19 one time whereas 25 two times to the next steps

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5.1 Crossover

String No.	Mating Pool	Crossover Point	Offspring after crossover	X Value	Fitness $f(x) = x^2$
1	01100	4	01101	13	169
2	11001		11000	24	576
3	11001	2	11011	27	729
4	10011		10001	17	289
Sum					1763
Average					440.75
Maximum					729

- Here Crossover points are randomly selected as 4 and 2 respectively. Before crossover point no change and after crossover swap bits between chromosomes
- Here Max. fitness value=729 which is better than previous max. value 629. Thereby improving the fitness of the offsprings.

5.2 Mutation

String No.	Offspring after crossover	Mutation Chromosome for flipping	Offspring after mutation	X Value	Fitness $f(x) = x^2$
1	01101	10000	11101	29	841
2	11000	00000	11000	24	576
3	11011	00000	11011	27	729
4	10001	00101	10100	20	400
Sum					2546
Average					636.5
Maximum					841

- Mutation Chromosome for Flipping (MCF) are selected randomly.
- Change the bit of Offspring Chromosome when MCF is 1 otherwise unchanged
- After mutation the max fitness value is increased to 841 better than 729

Step-6 Select chromosomes for the next loop based on actual count and Repeat the process until we get the max fitness value that is lower than the previous one.

End of Chapter